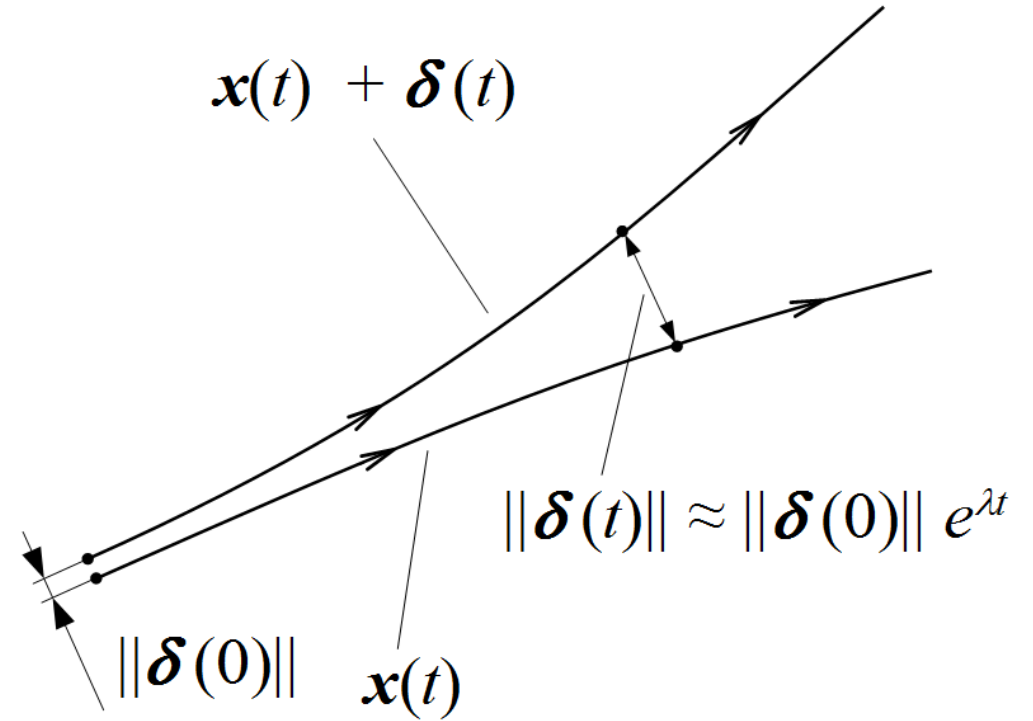


Day 27 - Hallmarks of Chaos

Conceptualizing the Lyapunov Exponent

*Trajectories diverge exponentially
in time*



Hallmarks of a Classically Chaotic System

1. **Deterministic**
2. **Sensitive to Initial Conditions**
3. **Non-periodic Behavior**
4. **Strange Attractors**
5. **Parameter Sensitivity**
6. (Sometimes) **Periodic Behavior**

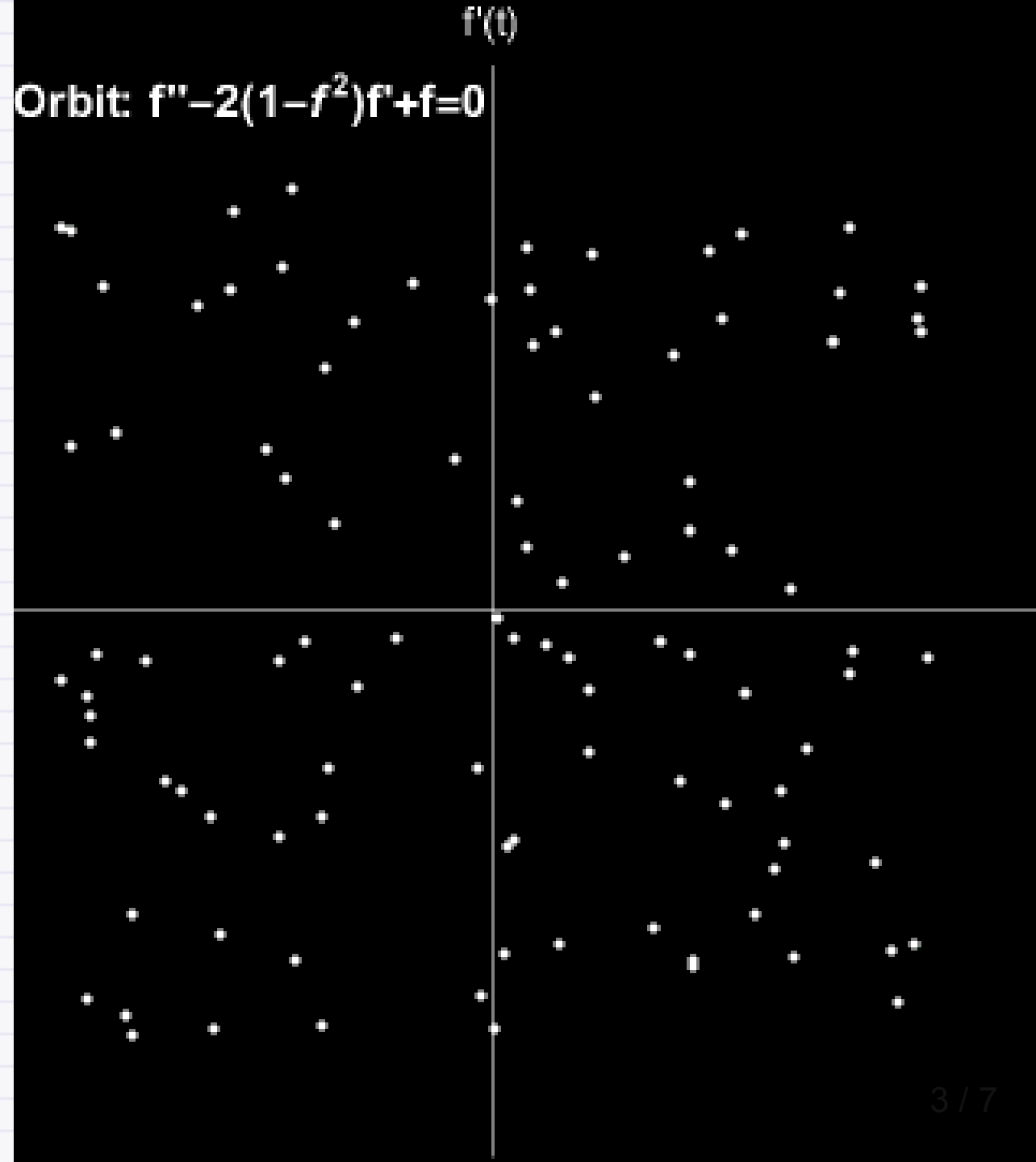
Limit Cycle

A **limit cycle** is a closed trajectory in phase space that is an attractor for a dynamical system.

The **Van der Pol Oscillator** exhibits a limit cycle.

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$$

Random initial conditions converge to a limit cycle. Modeled with $\mu = 2$.



The Lyapunov Exponent

$\vec{\delta}(t)$ is the separation vector between two trajectories in phase space $\vec{\delta}(t) = \vec{x}_2(t) - \vec{x}_1(t)$.

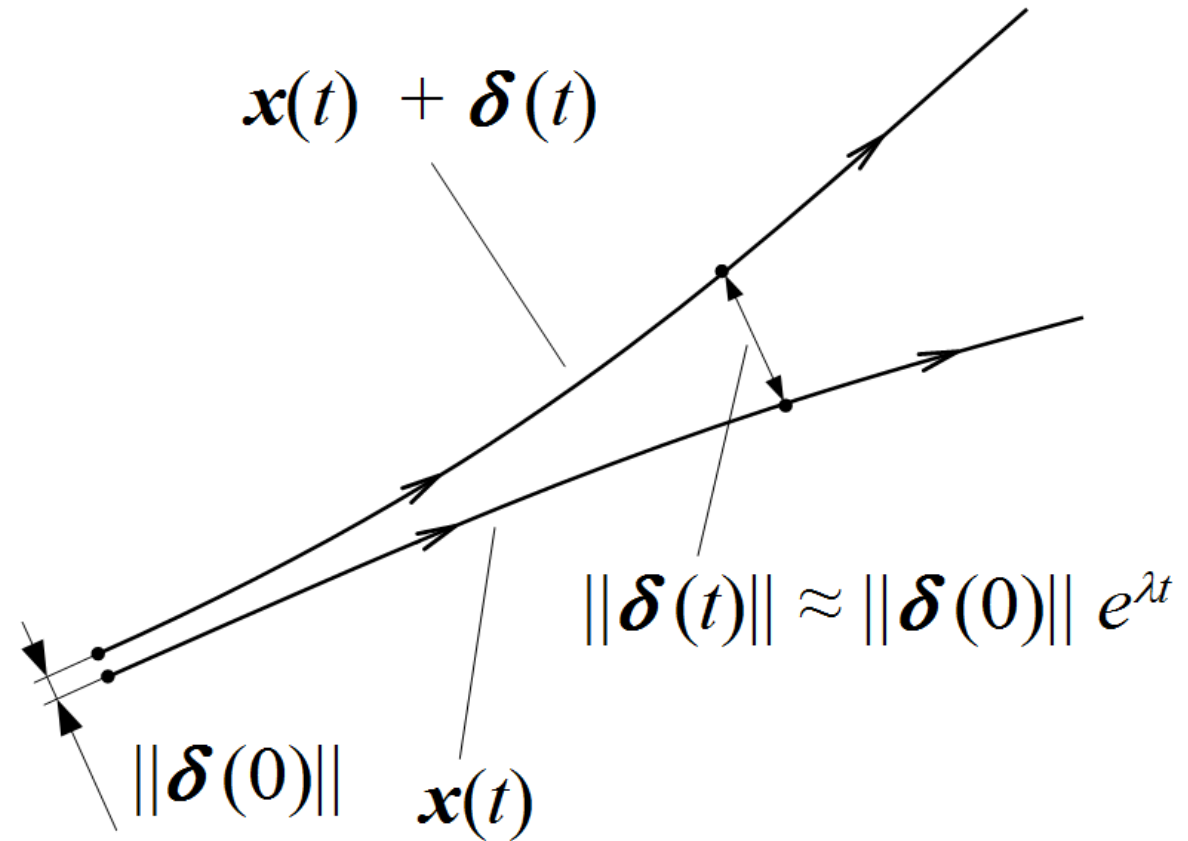
Do trajectories diverge exponentially in time,

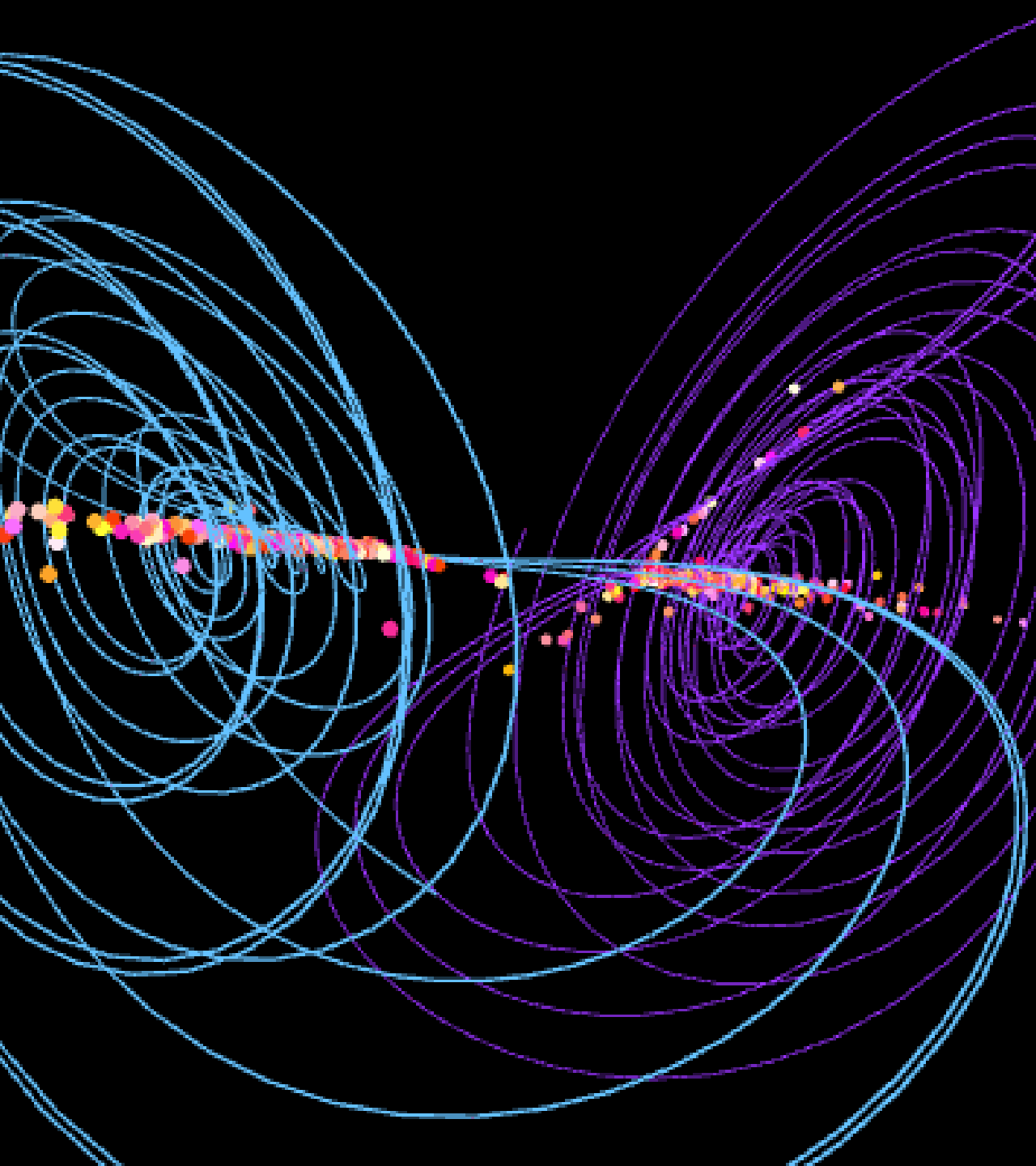
$$|\vec{\delta}(t)| \approx |\vec{\delta}(0)| e^{\lambda t}?$$

Each phase coordinate can change at a different rate:

$$\vec{\lambda} = \langle \lambda_1, \lambda_2, \dots, \lambda_n \rangle.$$

Largest $\lambda_i > 0$? Chaotic system.





Strange Attractors

A **strange attractor** is a set of points in phase space that a chaotic system approaches.

Chen Attractor

$$\dot{x} = \alpha x - yz$$

$$\dot{y} = \beta y + xz$$

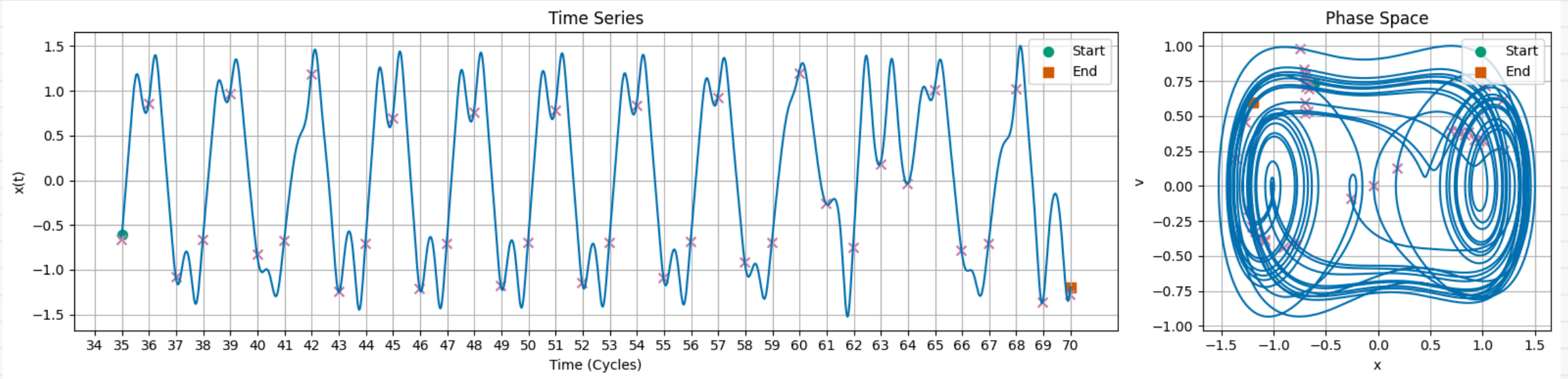
$$\dot{z} = \gamma z + xy/3$$

$$\alpha = 5, \beta = -10, \gamma = -0.38.$$

[Interactive 3D Model](#)

Example 1: Duffing Equation

$$\ddot{x} + \beta\dot{x} + \alpha x + \gamma x^3 = F_0 \cos(\omega t)$$



Exhibits Periodic and Chaotic Behavior

Illustrates period doubling bifurcations as route to chaos

Example 2: Lorenz System

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= x(\rho - z) - y \\ \dot{z} &= xy - \beta z\end{aligned}$$

**Exhibits sensitive dependence
on initial conditions**
**Demonstrates the concept of a
strange attractor**

